CHIME System Performance at Critical Design Review: A parametric model for end-to-end simulation analysis

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ABSTRACT

The Copernicus Hyperspectral Imaging Mission for the Environment (CHIME) will deliver routine hyperspectral observations over land and coastal zones in support of EU and related policies for natural resource management. In preparation for data exploitation, Sentinel User Preparation studies are ongoing in which CHIME features as a priority dataset for future use cases.

User preparation activities have traditionally relied on space-segment requirements specifications. While feasible, this approach tends toward conservatism and lacks the granularity needed by application developers — for example, SNR is typically provided only at a few reference radiance levels rather than across the full instrument dynamic range. To improve the fidelity of CHIME user preparation, we have developed dedicated parametric models of the system performance, anchored on the System CDR design status.

This contribution presents the motivation for the modelling approach and the models themselves, covering spectral, spatial, and radiometric performance domains. We will also discuss how these models can be exploited by application studies, their level of maturity at the current stage in the development, and any expected future evolution.

Keywords: System Performance, CHIME, User Preparation Analysis, SUP E2E analysis, Copernicus

1. INTRODUCTION

The Copernicus Programme, formerly known as Global Monitoring for Environment and Security (GMES) initiative, was established by the European Union to provide accurate and timely Earth Observation data in support of environmental management, climate change mitigation, and civil security. Its operational space component includes the Sentinel satellite missions developed by the European Space Agency (ESA). Growing user demand and new observation requirements led to the definition of the so-called Copernicus Expansion Missions or CopEx. The CopEx program fits between the first-generation Sentinel missions, currently operational, and the next-generation Sentinel missions currently under early development.

One of these six expansion missions is CHIME (Copernicus Hyperspectral Imaging Mission for the Environment), which will deliver routine hyperspectral observations to support sustainable agriculture, raw material management, food security, biodiversity monitoring, and soil and mineral assessment [1]. CHIME's payload is an advanced imaging spectrometer covering the visible to shortwave infrared spectral range with numerous contiguous bands, enabling the retrieval of detailed vegetation, soil, and mineral properties [2].

After the successful System Critical Design Review (CDR) performed in 2025, CHIME entered in Phase D, marking the beginning of system assembly, integration, and verification activities. Instrument and Platform Assembly, Integration and Testing (AIT) activities are currently underway, and the Instrument is expected to undergo a comprehensive performance characterisation campaign during thermal vacuum testing by end of 2028.

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As the mission progresses towards launch, user preparation activities have also been initiated to ensure the earliest possible exploitation of CHIME observations. Drawing on lessons learned from previous operational and scientific Earth observation missions, ESA has placed increasing emphasis on early and targeted user preparation. Investing effort early enough in these activities can secure a better data uptake and application usage since the beginning of the operational mission.

Within this context, the Sentinel User Preparation (SUP) initiative, implemented in ESA's FutureEO programme, aims to ensure that European researchers, service providers, and institutional stakeholders are ready to exploit the opportunities offered by the forthcoming Copernicus Sentinel Expansion and Next Generation missions. SUP promotes the development and validation of innovative applications, algorithms, processing methodologies, and user-driven services before mission launch, thereby reducing the gap between mission availability and operational uptake. Through a multi-mission approach, the initiative brings together scientific experts, industry, public authorities, and end-users to develop representative datasets, test emerging applications, and build the expertise required for rapid adoption of new Earth observation capabilities across a wide range of environmental and societal domains.



Figure 1. Sentinel's User Preparation initiative. The initiative combines Sentinels Next Generation and Copernicus Expansion future users. A first workshop is planned on November 2026 in ESRIN.

For CHIME, the availability of a realistic performance model for user preparation activities and applications development is an essential enabler. Since operational CHIME observations are not yet available, representative datasets must be generated using simulations that accurately reproduce the expected spectral, spatial, radiometric, and temporal characteristics of the mission. Such performance modelling allows researchers and application developers to evaluate retrieval algorithms, quantify expected information content, assess uncertainties, and determine whether mission requirements can be met for specific use cases. This is particularly important for applications that rely on subtle spectral signatures, including vegetation trait retrieval, crop condition and yield assessment, soil characterization, biodiversity monitoring, mineral exploration, and water quality monitoring. The performance of these applications is directly linked to the sensor characteristics. Spectral range and spectral resolution determine the ability to resolve narrow absorption and reflectance features associated with plant biochemistry, minerals, and aquatic constituents; radiometric performance and signal-to-noise ratio govern the detectability of weak spectral signals; spatial resolution influences the representation of heterogeneous landscapes and inland and coastal water bodies; while revisit frequency determines the capability to monitor dynamic processes and seasonal variability. Consequently, realistic modelling of the instrument performance, including spectral sampling, spectral response function, radiometric noise, spatial resolution, and temporal sampling, is required to assess application readiness and retrieval robustness before launch, supporting the development of mature and mission-ready applications prior to mission deployment.

Traditionally, visibility to the users has been somewhat limited before operations. Performance is shared in terms of the mission requirement documents and other publicly available data. The main issue with this approach is that requirements cover key aspects of the performance but do not provide the full picture that is required for application analysis which can lead to the wrong extrapolation. On the opposite end of complexity, dedicated test datasets can be generated for users based on the programme's end to end simulator capabilities. Whilst this would be completely performance representative, it is impractical to anticipate every scenario of interest to the user community. A decentralised approach where users can model the performance themselves is therefore more adequate. Besides these constraints, performance reporting at the current programme stage has non-negligible limitations: industry models are proprietary, right level of margins need to be considered as the system is being built and the right level of detail needs to be injected in the model whilst remaining agile and allowing parametric analysis.

To address these challenges the CHIME team has adopted an intermediate approach defining a simplified set of parametric models that cover all relevant system performances (i.e. radiometric, spectral, spatial). These models operate on at-sensor radiance inputs and generate representative Level-1 spectral radiance simulated data, reproducing the expected instrument response while remaining accessible to the user community. We consider that the approach sets a good precedent in terms of how programme and user communication should be at this stage, compatible with the programme constraints, but sharing significantly more information than typically carried out in the past. We hope this will allow users preparing and running their algorithms with representative data and decentralising the efforts on the assessment by sharing the modelling.

The remainder of this paper is organised as follows. In Section 2 we summarise the main performance capabilities of the CHIME mission and justify the modelling approach taken. Then in the following sections we cover radiometric, spectral and spatial performances respectively. Finally, we propose how the models should be integrated by the users if end-to-end (E2E) analysis are required and point to the GitHub repository that will host the models and we conclude with our suggested next steps.

2. CHIME PERFORMANCE MODELLING OVERVIEW AND PROPOSED MODEL SCOPE

The CHIME performance requirements derived from the user needs are specified in the CHIME Mission Requirements Document [3]. From these mission level requirements, a full space segment apportionment is covered in the Space Segment Requirement Document or SSRD which is the main applicable set of requirements to the CHIME industrial prime contractor.

Throughout the design and development of the mission, a range of dedicated performance analysis tools has been developed within the CHIME mission to support system design, performance trade-offs, performance monitoring, and compliance assessment against mission requirements. These tools provide the technical basis for evaluating the expected mission performance and support the verification process.

2.1 Performance breakdown and subsystem analysis

The first step in performance analysis is the complete breakdown of the system performance into lower-level contributions. This approach serves two main purposes: First, it enables the allocation of directly applicable performance requirements to the different subsystem suppliers. Second, it allows individual performance contributors to be analysed independently and in greater detail.

High fidelity local analytical models are developed as part of this task. One example is the Attitude and Orbit Control Subsystem (AOCS) pointing knowledge simulations. This model evaluates the pointing requirements compliance by the development of a complex dynamics simulator and Monte Carlo analysis. Another example is the optical analysis, which evaluates the final optical quality of the instrument taking into account manufacturing and integration tolerances.

These different performance tools are developed by CHIME industrial stakeholders and their outputs are all integrated in performance budgets managed by Instrument and System industrial primes. A status at CDR of this performance has been presented in [2][4]. The final performance verification and requirement closeout takes all these elements into account and demonstrates performance compliance by a combination of test and analysis.

By definition the scope of the tools and verification is directly tailored to the requirements conditions. For instance, the absolute radiometric accuracy of the mission is evaluated over specific synthetic scenes defined as part of the requirement. Similarly, the NeDL/SNR requirements are only evaluated for the top of atmosphere reference radiance levels.

This approach is fully consistent with the traditional systems engineering V-cycle and provides a robust framework to ensure that the mission meets its performance objectives. However, these tools and budgets are not directly suitable for supporting user application studies: most of the detailed performance models are proprietary to industry, while the performance budgets and verification scenarios are primarily focused on requirement-specific and worst-case conditions rather than representative operational use cases.

2.2 End-to-end image based simulators. Strengths and limitations

Complementary to the performance break-down approach, CHIME has developed E2E simulation capabilities. The Observation Performance Simulator (OPSI) takes as input oversampled Top of Atmosphere spectral radiance scenes, and outputs L1B and L1C TOA simulated products using the ground processor prototype (GPP) and implementing an instrument forward model that simulates with high fidelity the Instrument performance. The CHIME End to End Simulator or CHEES [6], integrates the OPSI into a full end to end pipeline that takes as input Bottom of Atmosphere reflectance and outputs L2a CHIME like products simulating the full space segment and ground processing pipelines.

The role of these simulators is not to substitute but to complement the traditional verification approach described in the previous section. Being image-based simulators with heavy processing demands, they are not suited to statistical or Monte Carlo analysis that the complete verification process requires. However, they complement the verification in three fronts:

- The prototype processors can expose residual performance errors that the analytical error breakdown might have missed
- As they operate at image level, they can reveal how the mission performs on scenes representative of the true geophysical variables, as opposed to the corner cases captured by requirements (e.g. straylight residual over real scenes vs synthetic cases, or residual striping over real dark data instead of dark synthetic cases).
- They allow understanding the impact of Non-Compliances or held margin on the final product.

A simulator of these characteristics could be very well suited for user application analysis, and this is indeed planned as a potential use case for it. It provides the capability of simulating from Bottom of atmosphere to L2 if that's desired, using CHEES; alternatively, the sole OPSI component can be run if local atmospheric correction techniques are being traded.

Two possible approaches were considered to provide image-based simulations results to the SUP initiative:

- Share the current CHEES/OPSI simulator infrastructure with specific users within the SUP initiative
- Executing user-defined scenarios at ESA premises and delivering the resulting data products.

Both of the options were discarded to support the on-going SUP studies, facing clear drawbacks.

OPSI and CHEES, both used for a significant number of use cases for prototype demonstration to performance verification and scientific assessment, are very high fidelity but not simple to use by a more general user and their configuration requires in depth knowledge of the CHIME Instrument. Moreover, the simulators are still under development and currently under validation before the Science Readiness Level 6 assessment for CHIME. Sharing them in their current form was not an option.

The alternative of performing simulator runs on demand to support users, creates an artificial dependency between the users and ESA, which could lead to unnecessary bottlenecks and require a dedicated simulation team.

2.3 A dedicated simplified parametric model for user application analysis

The limitations of both traditional performance budgets and high-fidelity end-to-end simulators motivated the development of a complementary performance modelling approach. The goal was to take all the required elements from the performance breakdown and budgets and the high-fidelity E2E simulators and produce a simplified parametric model simple to run and ingest in simulation pipelines but that would capture the Instrument behaviour with the right level of fidelity.

Unlike the detailed industrial models, the proposed approach can be readily shared with external users and integrated into third-party simulation environments. At the same time, it avoids the complexity, computational cost, and operational constraints associated with the high-fidelity simulation framework.

The model is therefore not intended to replace either the industrial verification tools or the end-to-end simulators. Instead, it provides a lightweight and efficient means of simulating CHIME performance, supporting user application studies, sensitivity analyses, and mission exploitation activities, while maintaining consistency with the latest system-level understanding of the mission performance.

3. RADIOMETRIC PERFORMANCE

3.1 Signal to Noise Ratio (SNR) and Noise Equivalent Delta Radiance (NEDL)

Context

SNR / NEDL performance as in many other spaceborne optical missions is specified in CHIME for a given reference at-sensor Top of Atmosphere spectral radiance profile or Lref, tabulated values provided in Appendix. The CHIME reference radiance was defined for a standard atmosphere, a biome spectrally flat reflectance of 20% and an observation geometry at nadir with a Sun Zenith Angle or SZA of 30 degrees [3].

While performances at Lref provide a useful reference point, user application analyses require the understanding of sensor performances across the full dynamic range. The CHIME dynamic range is defined in the Space Segment specification for a set of spectral radiance curves defined in the Mission Requirements Document [3] based on the variability of observation geometry (e.g. SZA variation along the orbit) and covering a wide range of reflectance cases (ranging between 3% up to 100% biome reflectance).

Application studies are typically based on the simulation of real scenes where the measured dynamic can be significantly different than what is captured at Lref. Consequently, the use of a single SNR value or any derived approximation might lead to underestimation or overestimation of the retrieval performance potentially leading to misleading conclusions. A parametric SNR model based on the incoming at-sensor spectral radiance therefore provides significant added value for performance assessment and application-oriented analyses.

The instrument SNR as a function of spectral radiance is predicted by the CHIME Instrument Prime Contractor using a detailed instrument model that accounts for the complete optical chain and detector characteristics. Although this model cannot be distributed, its behaviour can be approximated using a simple two-parameter representation.

This simplification is possible because the instrument noise regime is bounded by two dominant mechanisms: detector readout noise at low signal levels and photon shot noise at high signal levels. By fitting two parameters independently for each spectral channel, the complete SNR-versus-radiance relationship can be reproduced over the mission radiometric range.

Model Description and fit justification

The model is provided in the equation below. The g and N0 parameters are fitted per channel and provided in the appendix for the performance status as per the System CDR.

$$SNR(\lambda) = \frac{L(\lambda)}{Lref(\lambda)} g(\lambda) \frac{1}{\sqrt{\frac{L(\lambda)}{Lref(\lambda)} + N0(\lambda)}} \qquad (1)$$

NEDL as per the definition can be then estimated from the SNR per channel as follows:

$$NEDL(\lambda) = \frac{L(\lambda)}{SNR(\lambda)} \quad (2)$$

The two fit parameter holds as long as the dominant noise sources can be described as a combination of a proportional to the square root of the signal (e.g. shot noise) and a fixed offset (e.g. readout noise). Any term that is proportional to the signal (e.g. flicker noise, detector 1/f noise, detector gain variability) is not covered by this model but in the case of CHIME these terms are second order, assessment that is backed by the Industrial Prime analysis and the adequate fit reported later, see Figure 2.

The fitted model has been compared to the Industrial contractor model. As depicted below, the fit maps well the predicted SNR over the full dynamic range.

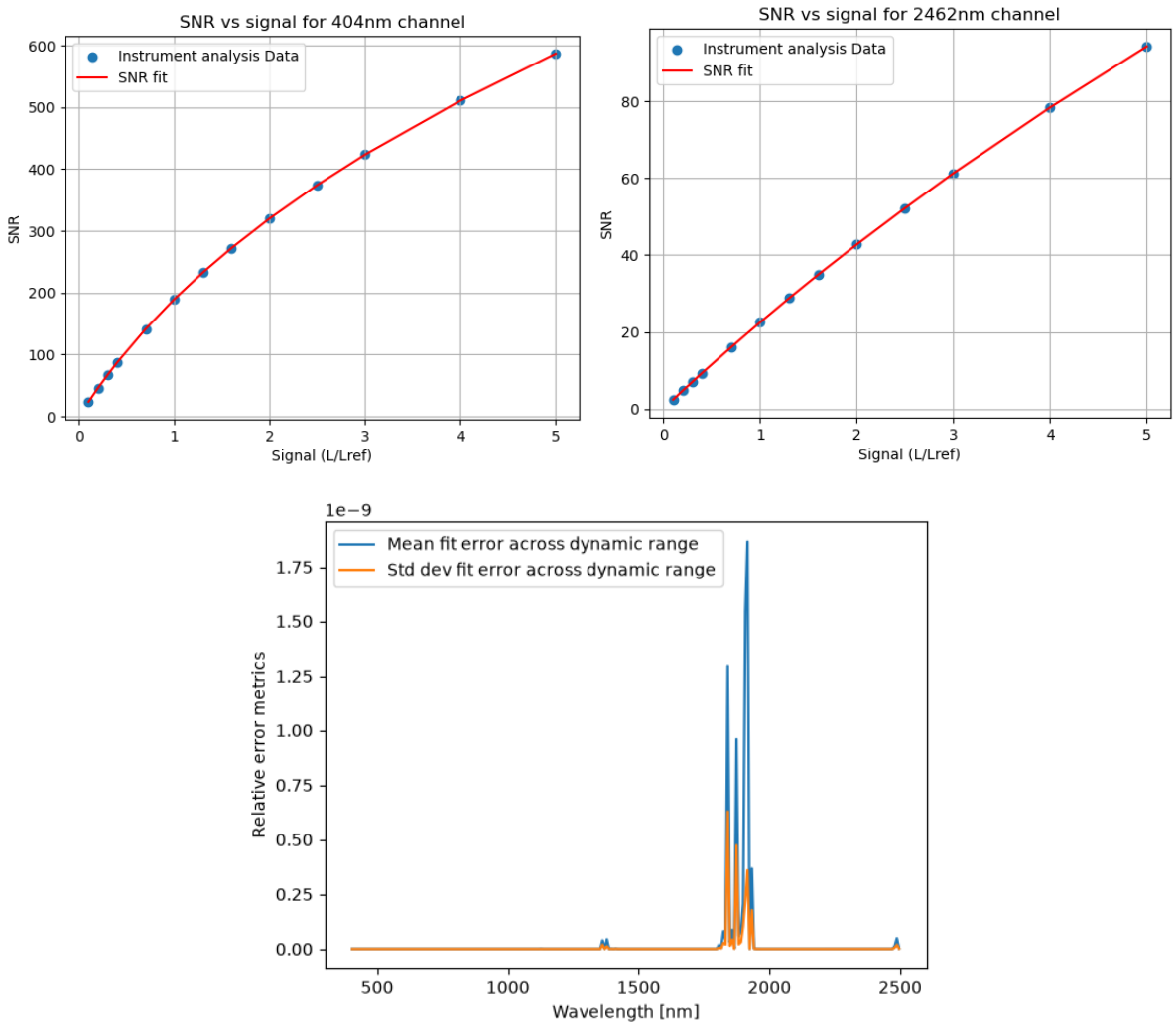


Figure 2 SNR model parametric fit (red) comparison with detailed Industrial partner performance model (blue) for two spectral channels across the dynamic range (top). Fit relative mean and standard deviation error for all spectral channels over the dynamic range (bottom).

The model can be used to predict the Instrument SNR over any point in the dynamic range. Some examples are shown below for different radiance ratios to Lref (L/Lref cases):

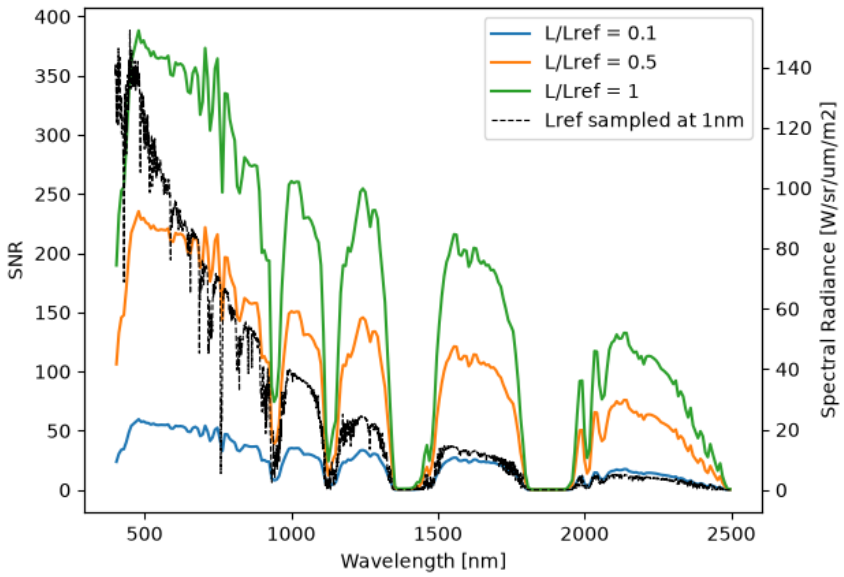


Figure 3. SNR model parametric fit results for different at-sensor radiance levels across the spectral range of CHIME

Model validity and expected evolutions

The model parameters provided in the Appendix have been fitted for the Instrument performances at System CDR. The performance relies on several subsystems parametrization and subcontractors requirements and commitments. Below we list the key main assumptions and performance drivers.

- Telescope transmission, based on mirror reflectivity
- Spectrometer throughput, including grating diffraction efficiency and spectrometer mirror reflectivity
- Detector characteristics, including dark signal modelling, QE and noise characteristics
- Provisions for EOL degradations

As discussed in the introduction, the CHIME project is now entering in Phase D and Instrument subsystems completion and integration is under way. As part of the normal lifecycle of the project, several of these subsystems have now actual performances and not allocations or supplier commitments.

The following plot is a prediction at instrument level considering the latest measured performance from different subsystem flight candidates. The resulting performance is better than the original commitment resulting in higher SNR performance than the requirement for most of the spectral channels, see below.

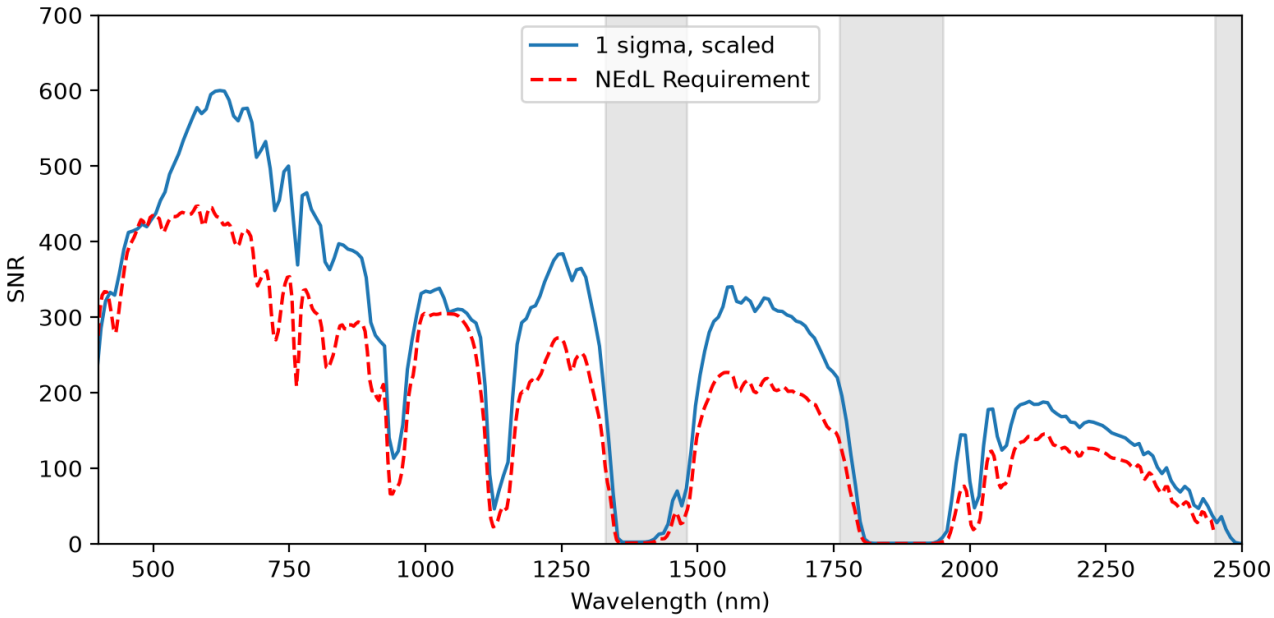


Figure 4. Latest performance predictions including different subsystems characterised performance vs requirement.

The CHIME team will keep monitoring the subsystem development throughout the system integration activities and issue updated parameters after the Instrument Performance Characterisation in TVAC. Until then, the current fit provided is considered an adequate envelope for this performance as it's more informative than a simple requirement definition and envelopes well the current predicted performances.

3.2 Absolute Radiometric Accuracy over uniform scenes

3.2.1 Context

The radiometric accuracy of CHIME is specified by requirements over uniform scenes. This is a standard practice in Earth Observation although recently new missions are moving to a total uncertainty specification. In order to keep traceability with the specified requirements we stick to the radiometric accuracy definition and understanding as per the MRD [3].

We define radiometric accuracy as the radiometric error of the instrument when performing a measurement for a duration long enough so that temporal noise can be averaged out or is second order. The residual error of such measurement is the accuracy of the Instrument. Typically to model this we simulate acquisitions over synthetically homogenous scenes where the noise of the instrument is averaged out.

The Instrument radiometric accuracy is normally specified over a reference radiance (Lref) and performance over the dynamic range is not provided. Typically, this is defined in percentage terms for a given confidence level. In CHIME the requirement is 4% for a confidence level of 68% at Lref.

For a mission like CHIME the accuracy can be classified as:

- Errors in the gain used for the calibration. This can either be due to an error in the gain calibration or an evolution of the instrument behaviour since the last calibration
- Errors in the offset calibration. This can either be due to an error in the offset calibration or an evolution of the instrument behaviour since the last calibration.

Consequently the accuracy behaviour can be modelled as follows:

$$ARA(\lambda) = k(\lambda) + \frac{L_0(\lambda)}{L(\lambda)} \quad (3)$$

where k is the error in the gain, L_0 the error in the offset, and L the at sensor radiance. The linear sum is justified in CHIME case as the main drivers of the gain and offset terms are of very different temporal nature. Gain calibration is planned at a weekly rate but its main error contribution comes from the knowledge error of the diffuser reflectance, which is a bias, whereas offset calibration is foreseen before and after land acquisitions and hence contributes as a standard deviation. As such the error best approximation is a linear sum instead of a quadratic sum. The offset error injects a per pixel at sensor radiance dependence, which prevents us from using a constant relative value regardless of the radiance measured.

The proposed model and fit parameters are a conservative signal-dependent performance error estimation. They do not substitute a thorough uncertainty propagation but provide a first order modelling of the radiometric accuracy instead.

3.2.2 Model Description and fit justification

We propose a four-parameter model to capture this behaviour. This model sits in between two common practices. On the one hand many missions provide radiometric accuracy as a single percentage value at reference level. Whilst this can be a good first order approximation it significantly fails at low radiances, where the offset term in equation (3) becomes dominant. On the other hand, per pixel uncertainty products are starting to be provided in multiple EO missions (e.g. Sentinel-2 RUT or Sentinel-3 [7][8]). However, these are full-fledged propagation algorithms that escape the scope of this parametric exercise; CHIME will also provide per pixel uncertainties, its pipeline is currently under development. The proposed model sits in between these two options and provides the right level of granularity for applications assessment over the whole dynamic range:

$$ARA(\lambda) = k + \frac{L_0(\lambda)}{L(\lambda)} = k + \frac{S_0[e-]}{S(\lambda)[e-]} = k + \frac{S_0[e-]}{a(\lambda)+b(\lambda)(\frac{L(\lambda)}{L_{ref}(\lambda)})} \quad (4)$$

The model simply takes Eq. 3 but rewrites taking into account the following considerations:

- The gain error is approximated constant over the spectral range by parameter k. The current model relies on apportionments to multiple contributions with the driving performances coming from the on-ground diffuser BRDF measurements and the Lunar calibration correction. A flat value of 2.5% is considered enveloping and compatible with the parent requirement based on Industrial contractor assessment.
- The offset error contribution is based on an estimate of the offset in electrons captured by S0 parameter over the equivalent electrons to the incoming TOA radiance (labelled as S in Equation 4) which is parametrized per channel using the a and b parameters.

The current fit parameters are provided in the table below:

Table 1 ARA Performance parameter fit

k	**S0**	**a**	**b**
0.025	150	Provided in appendix per channel	

Below we show the ARA estimated performance from the current fit for different radiance levels. At Lref the performance is compliant with the requirements besides the agreed exclusion regions (e.g. atmospheric absorption windows and far SWIR flagged in grey).

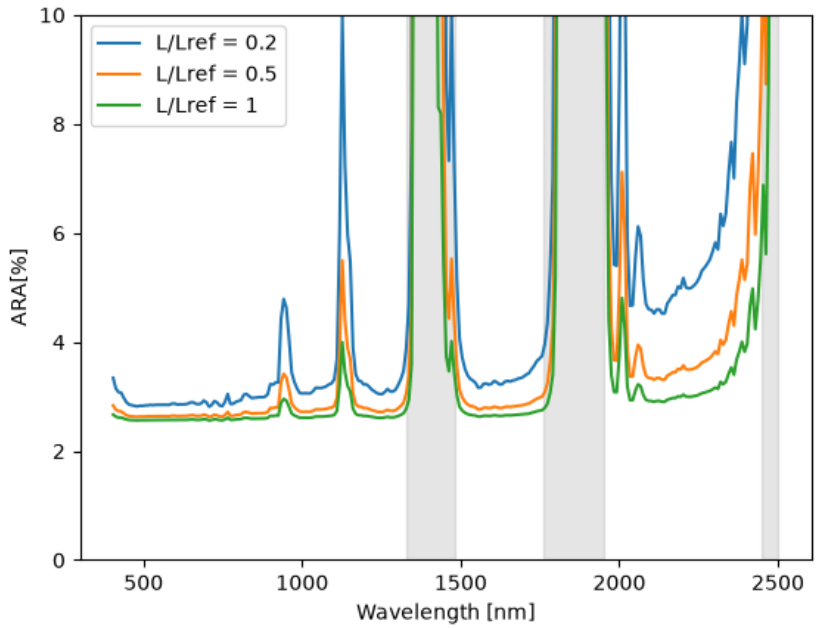


Figure 5. ARA parametric model results vs different levels of at-sensor radiance normalised to Lref. The greyed areas are excluded from the requirement evaluation, being regions of high atmospheric absorption.

3.2.3 Extra considerations over heterogeneous scenes

The radiometric accuracy modelling just described applies over uniform scenes or for per pixel clusters homogenous enough so that there is no contamination across pixels neither in small or large spatial scales along the scene.

Application analyses are likely to be run over heterogeneous scenes that could exhibit high radiometric contrast in some cases. Below we provide specific considerations to take into account the extra radiometric error that could be faced in such situations.

Extra radiometric error can be induced in CHIME mainly by three phenomena: Straylight, PSF tail effects and geolocation error.

Straylight

CHIME straylight is controlled by specific requirements that are made applicable for a knife edge scene defined for the extremes of the dynamic range, see figure below:

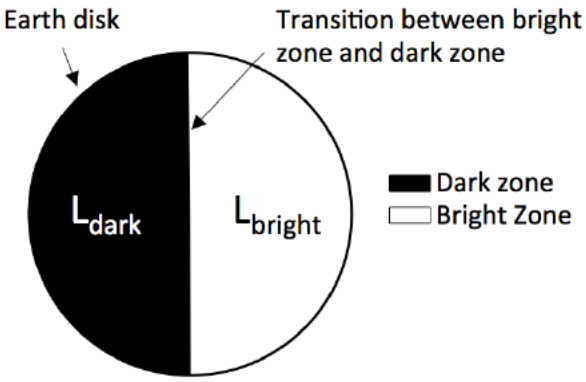


Figure 6. Non-uniform scene defined in CHIME specification to control Straylight performance

The current performance after straylight correction can be approximated by increasing the parameter S0 from the baseline value of 150 to a range between 200-250. This is a very conservative estimation and should only be applied when there is significant contrast in the scene (e.g. bright sand vs dark water)

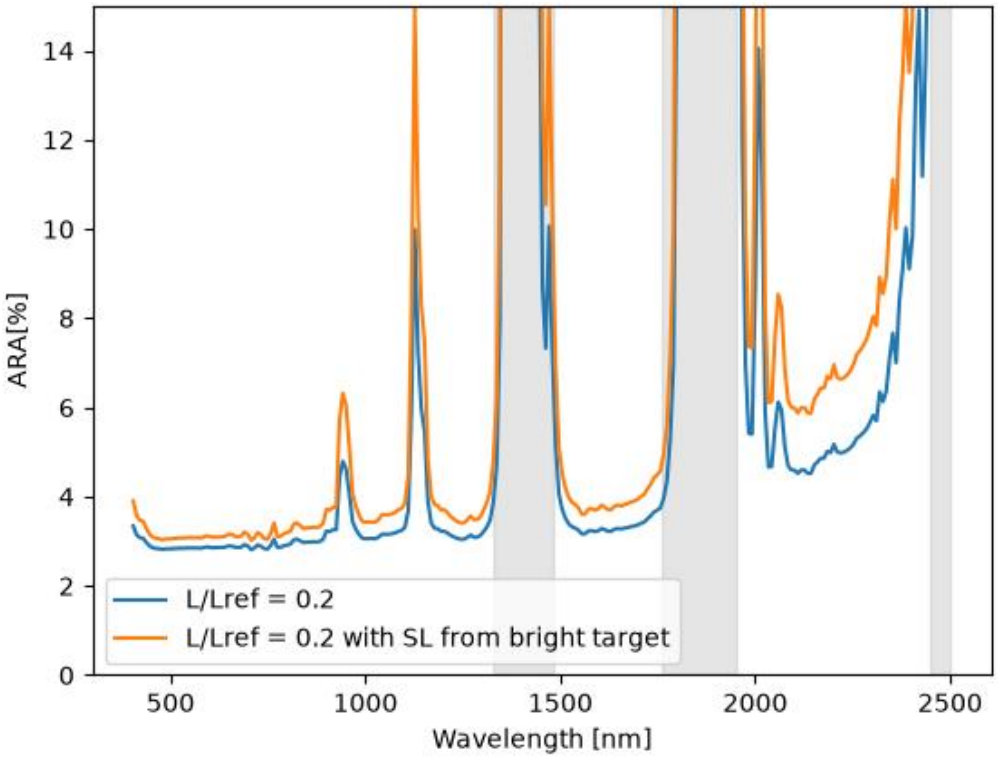


Figure 7. Comparison of ARA performance over uniform scenes vs extra degradation residual after correction due to non-uniform scene induced Straylight from a significant bright target (comparable to Lmax radiance). The greyed areas are excluded from the requirement evaluation being regions of high atmospheric absorption.

PSF tail effect induced radiometric error

The spatial resolution performance, discussed later in detail in Section 5.3, is defined as a Modulation Transfer Function (MTF) at Nyquist of 0.2-0.3. This is a typical performance figure in Earth Observation missions. This range secures a good balance between a too sharp MTF (which can induce aliasing) vs a too blurred MTF (which can cause loss of resolution). Still a consequence of this specification is that the adjacent pixels to the centre typically experience a radiometric leakage of ~10%, for the MTF range considered. Users do not normally take into account this phenomenon in their applications and treat each pixel as if it had been acquired by a perfect top-hat function, see Figure 8 and as such this leakage can propagate into a radiometric error.

The induced radiometric error caused by this tail leakage effect can be estimated as follows:

$$\Delta L = \sim \sum_n \frac{(L_n - L_{centr})}{L_{centr}} PSF_{adj_{weight}} \qquad (5)$$

where PSF_adj_weight can be approximated for the current performances to a value of 0.1 and n is the four adjacent pixels. Please note that this extra radiometric error should only be included in the assessment if a spatial convolution is not performed as part of the analysis. If the implementation suggested in this paper is considered (see Section 6 for further details) the addition of this error is done automatically by the convolution operation and as such its separate consideration is not required.

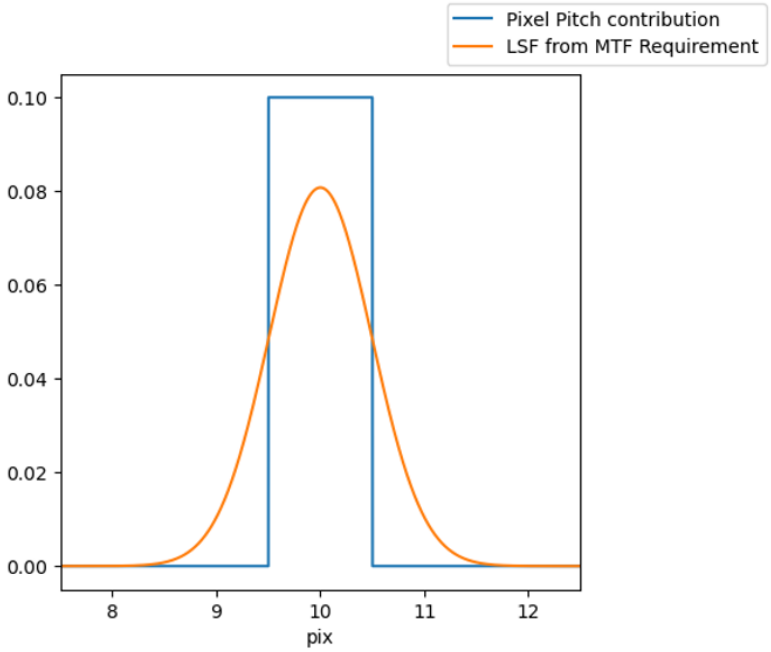


Figure 8. Comparison between Instrument specified PSF vs ideal pixel acquisition or top-hat. The current specified PSF can lead to a radiometric leakage of up to 10% to each adjacent pixel.

Radiometric error induced by Geolocation error

CHIME geolocation error is specified to 0.3 pixels at 95% confidence level w.r.t the Sentinel-2 Global Reference Image. The impact of geolocation error in the retrieval accuracy has been recently studied and modelled by an ESA in-house algorithm presented in SPIE [5] . A simple approximation for a unidimensional edge case of the induced extra radiometric error caused by geolocation can be approximated as follows:

$$\Delta L = \sim \frac{(L_{adj} - L_{centr})}{L_{centr}} \Delta gel_{pix} \qquad (6)$$

where Δgel_{pix} is the geolocation error at 1-sigma ~0.15 pixel and the radiance difference is the relative radiance contrast between adjacent and central pixel in the same fashion as per the PSF tail contribution. The algorithm to estimate uncertainty follows this principle but expanded to 2 dimensions, the reader is referred to the conference proceedings for further details [5].

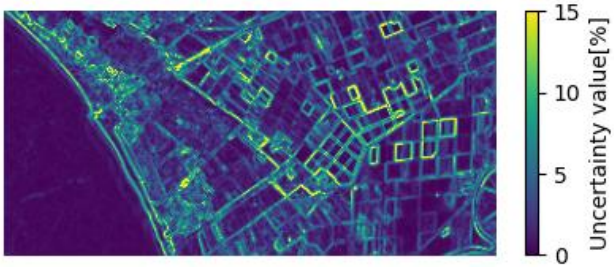


Figure 9. Additional radiometric uncertainty due to 0.3pix (2-sigma) geolocation error simulated from a Grosseto (Tuscany) Sentinel-2 655nm acquisition [5].

3.3 Relative Spatial Radiometric Accuracy

Besides Absolute Radiometric Accuracy, CHIME has a specific requirement on the relative spatial radiometric accuracy (RXRA) in order to control any residual striping in the product. The main contributions to the RXRA are the spatial equalization or flat-fielding residuals and the linearity correction residual.

CHIME calibration concept envisages a dedicated 90-degree yaw manoeuvre to acquire the same location with all pixels over a Pseudo-Invariant Calibration Site (PICS) and correct the residual equalization error [12]. The performance required is 1% at 68% confidence level. The currently expected performance is ~<0.4% for most channels. For channels with high spectral radiance relative gradient (e.g. atmospheric absorption regions), the performance can further degrade which is a limitation to this vicarious method. The equalization parameters update is nominally planned every 6 months and hence the error is expected to change at this characteristic timescale.

Our proposed modelling envelopes the current performances determined by dedicated high fidelity simulations of the site, whilst being simple to implement. The model allocates 0.4% residual error to all channels, but the ones listed in the Appendix. As shown in figure below, this parametrisation envelopes well the latest equalization residual analyses results at CDR based on simulated CHIME acquisitions over the baseline PIC site (Libya -1).

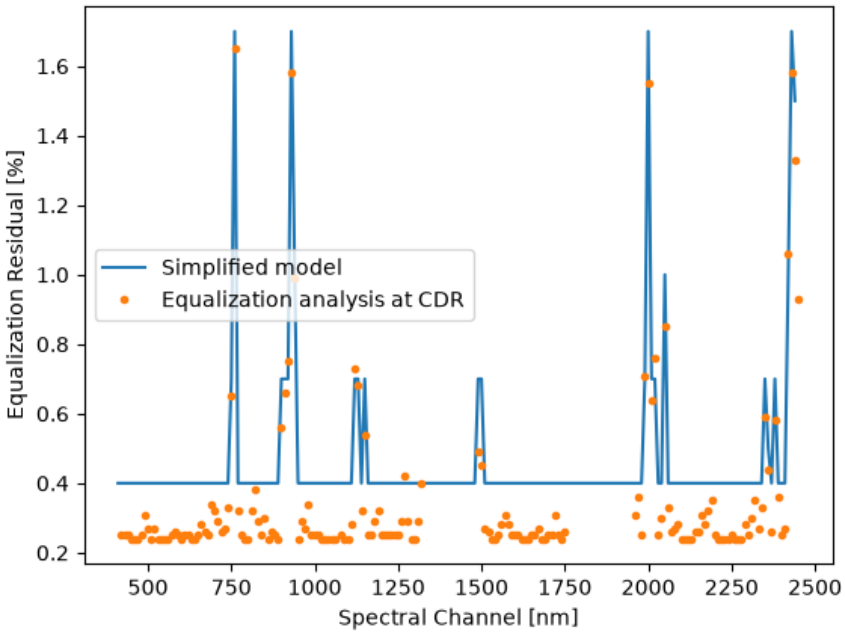


Figure 10. Comparison between simplified model and equalization residual analysis performed over simulated data with OPSI over Libya-1 PIC site at the time of System CDR. Courtesy of VITO Remote Sensing and OHB.

The implementation of the equalization residual error in the modelling can be only performed when working in the instrument geometry, as in this geometry the error just appears between columns for each spectral channel. In Section 6 we discuss in further detail how to simulate the instrument acquisition geometry from the orthorectified input.

For every channel k, and each spatial pixel column i the calibrated spectral radiance including equalisation error L_eq can be derived from the calibrated spectral radiance L as follows:

$$L(i,:,k)_{eq} = L(i,:,k) \cdot (1 + N(0,{\sigma^2}_{eq})) \qquad (7)$$

where $N(0,{\sigma^2}_{eq})$ is a gaussian random draw from a normal distribution centred in zero and with standard deviation equal to the equalization residual (i.e. allocation of 0.004 for all channels but the ones listed in the Appendix where the expected error is higher).

4. SPECTRAL PERFORMANCE

4.1 Spectral Response Function and Spectral Sampling

Context

CHIME spectral resolution and sampling are specified in terms of FWHM and SSI to less or equal than 10nm respectively for all channels.

Based on the Instrument optical design the as designed spectral sampling is 8.4nm. The expected variability of the sampling across the field is very low, far less than 1nm. As such the sampling does not need a dedicated model and a value of 8.4nm +/- 0.04nm can be used.

Regarding the spectral resolution, this is controlled by the Instrument Spectral Response Function. The spectral resolution performance is the combination of multiple elements within the optical system but it's mainly driven by the slit, the spectrometer and the detector. The actual shape is the combination of the response functions of each subsystem and even if the slit or the detector are top-hat functions, the final convolution into a single system function approximates well to a gaussian behaviour. This is not a new finding, the CHIME Product format specification is consistent with this assumption and will report ISRF curves based on central wavelength and FWHM following this gaussian approximation as well and this approach is also the baseline in other hyperspectral missions like EMIT, PRISMA or ENMAP [9][10][11].

Model Description and fit justification

For application analysis, users simply should take the estimate of the FWHM and convolve the input with a gaussian function consistent to it, using the formulas below.

$$\sigma_{ISRF}(\lambda_0, ACT) = \frac{FWHM(\lambda_0, ACT)}{2\sqrt{2\ln(2)}} \quad (8)$$

$$ISRF(\lambda, \lambda_0, ACT) = \frac{1}{\sigma_{ISRF}\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left[\left(\frac{\lambda-\lambda_0}{\sigma_{ISRF}}\right)^2\right]\right) \quad (9)$$

After the convolution the scene needs to be down-sampled to the Instrument grid. This depends on the relative sampling between the input scene and the CHIME spectral sampling. There is no need to include the degradation resulting from the detector in the down-sampling process as this is already included in the ISRF model.

The actual FWHM is variable across the field, leading to a function that has a wavelength and field position dependence. The overall behaviour depends on the final as built spectrometer performance however estimates from latest analyses performed for CDR indicate a variability between 9.7 to 10.7 nm across the field, see below:

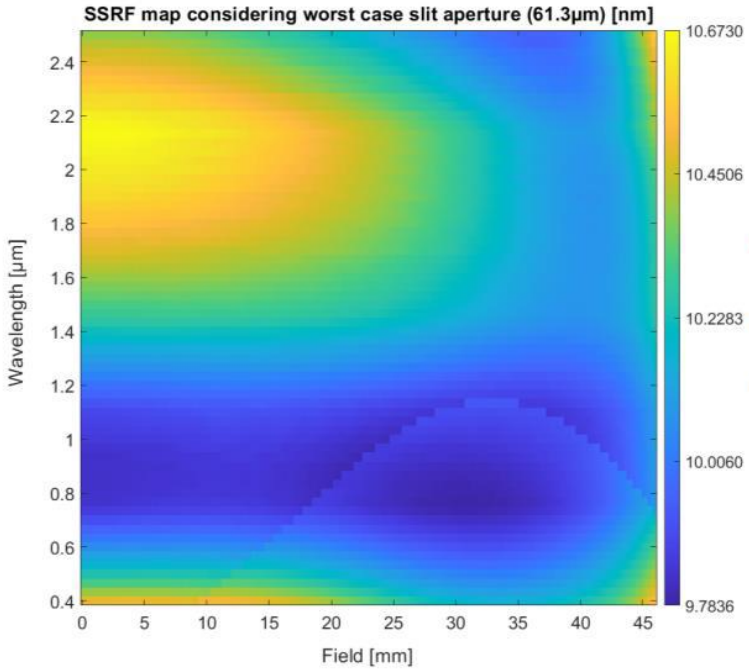


Figure 11. ISRF FWHM expected evolution for a single spectrometer across field and wavelength. Courtesy of AMOS/OHB.

This represents a deviation from the 10 nm specification and is currently subject to the applicable requirement deviation process. In order to use a representative user-modelling envelope, a range of 9.7–10.7 nm is proposed.

4.2 Spectral barycentre knowledge

Besides the spectral resolution and sampling, a key performance for the retrieval is the spectral barycentre knowledge. CHIME spectral knowledge along the mission lifetime is specified down to 0.5nm. The performance is secured thanks to the use of an on-board spectral calibration subsystem, whose design takes advantage of the Sun diffuser calibration required for the gain calibration. The envisaged mechanism, slots a doped filter glass window between the spectrometer and the Sun diffuser in a location close to the slit [12]. The doped filter has specific characteristic absorption features that are measured on-ground with high precision (0.1nm at 3σ), see figure below. Monitoring the absorption features measured by each spectrometer when looking at the Sun Diffuser via the doped glass, the spectral centre knowledge error per pixel can be controlled with a performance compatible with the requirement of 0.5nm.

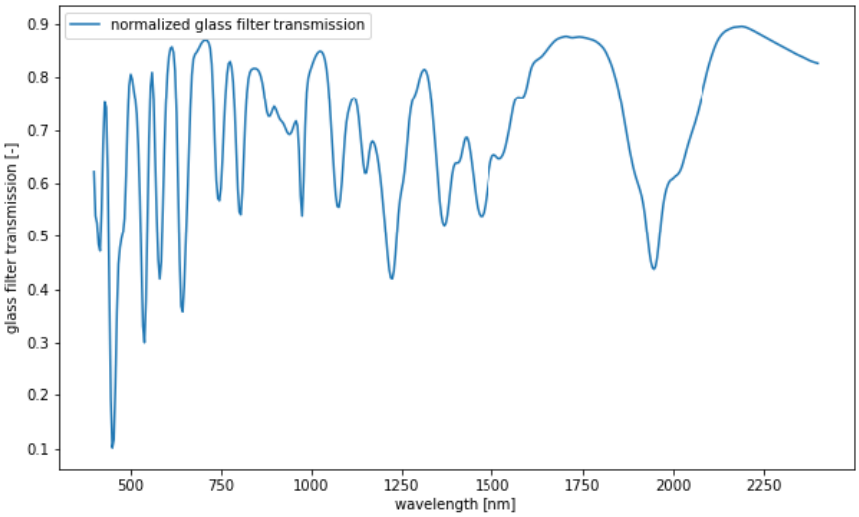


Figure 12. Transmission spectrum of the NIST SRM 2035b filter glass [12]

As CHIME spectrometer design acquires the full dynamic spectral range with the same grating and all channels are calibrated by the same doped filter glass, it is expected that the nature of the residual knowledge error is highly correlated across all spectral and field directions. However, until Instrument Performance Characterisation provides a better estimate of the correlation structure, the split of this 0.5 nm error between correlated and non-correlated components should be treated as a user sensitivity assumption rather than a CDR-predicted quantity. A correlation parameter f_c, ranging from 0 to 1, can be introduced to represent the fraction of the total error variance assigned to the correlated component, with the remaining fraction assigned to the non-correlated component. Users should study how the degree of correlation affects their application through sensitivity cases, for example $f_c = 0$, 0.5 and 1, until better knowledge of the performance is obtained by test and simulation.

4.3 Spectral misregistration

CHIME spectral co-registration error is specified as 0.1SSI or 1nm at Level 0, or without ground processing correction. This value secures that the contribution to the retrieval error of misregistration is second order and that smile correction is not required in the processing. The performance is met at spectrometer level (or sub-swath level), i.e. any pixel within a subswath meets currently this requirement, see Section 5.1 for further explanation on CHIME sub-swaths.

On the other hand, the performance between different spectrometers or sub-swaths can be further degraded to up to 0.5 SSI worst case, due to inter-alignment errors during integration. This is not a new phenomenon; it is a limitation observed in other instruments that meet acquisition requirements combining a suite of optical cameras (e.g. OLCI) and it's a limitation on the design and integration tolerances.

It's important to understand this caveat when processing pixels in areas that intersect two sub-swaths. What is fundamental is that CHIME product provides spectral barycentre knowledge per pixel with a performance of 0.5nm. As such the misregistration impact in applications can be mitigated if the right channel barycentre information is assigned to each pixel, which the current product format definition currently envisages.

5. SPATIAL AND GEOMETRIC PERFORMANCES

5.1 Observation Geometry

CHIME is required to meet a geometric revisit of 12.5 days or less [3].The implementation solution to meet this requirement is the use of two CHIME satellites orbiting in antiphase with a resultant flown-down required revisit for each of them of less than 25 days [2]. This drives the orbit design and the instrument swath to a total instrument swath of approximately 130km. To meet this performance the instrument makes use of three identical spectrometers with respective sub-swaths, see figure below. The spectrometers field of view are arranged securing a 30 pix or ~0.08° overlap in the ACT direction which is approximately 900 m at the equator for the baselined satellite orbit.

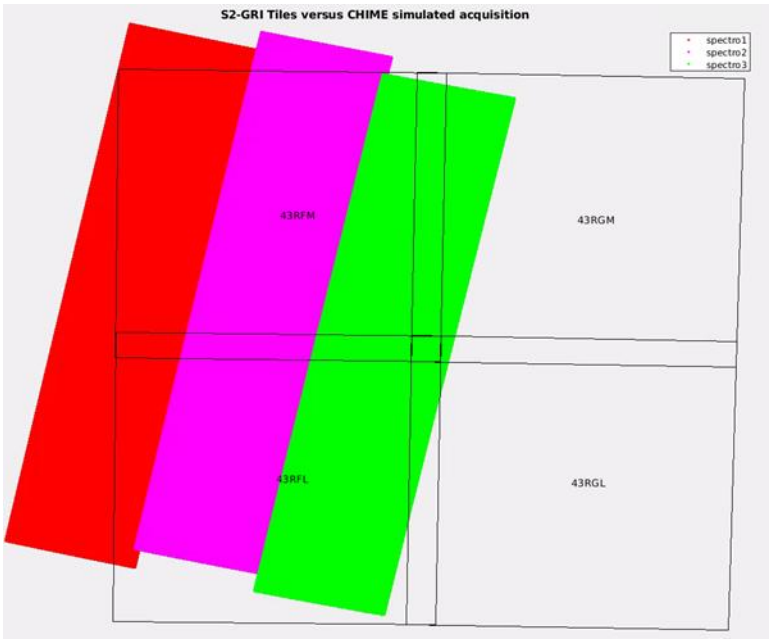


Figure 13. CHIME acquisition geometry. The full swath of ~130km at equator is achieved by the overlap of 3 identical spectrometers subswaths. Courtesy of Thales Alenia Space.

The three different sub-swaths (i.e. each spectrometer swath) are kept at L1b as separate products and stitched together as part of the orthorectification at either L1c or L2. The stitching process has been developed so that spatial mixing does not occur, i.e. pixels between spectrometers are not mixed and the stitching line is defined associating to each pixel solely one CHIME spectrometer.

5.2 Spatial sampling

The SSRD specifies an SSD for both L1b and L1c products smaller or equal to 30m. CHIME L1c products will be resampled into a consistent grid to the Sentinel-2 Global Reference Image (GRI) with a pixel sampling of 30m instead of the native 10m from the GRI. Being 30m a multiple of the Sentinel 2 GRI 10m pixels, full colocation is ensured. As such, for applications using directly orthorectified data, a sampling of 30m should be considered.

For applications using L1b, or for any other processing level that retains the Instrument geometry such as the L2S products, the Spatial Sampling Distance depends on the Instrument angular sampling or SSA, the Satellite geodetic altitude and the position in the swath. The variation between the pixel sampling distance at edge of ACT swath and nadir is given for the

entire CHIME (FOV~11.7º) in the table below. For reference the Instrument IFOV including tolerancing and lifetime effects is 45.85 urad, and the orbit reference average altitude is ~632 km as per the orbit design [1]. At maximum altitude and at the edge of the FoV, the predicted SSD locally reaches ~30.5 m; this small deviation from the 30 m requirement is covered by an accepted Request for Deviation.

Table 2 Variation of ACT Spatial Sampling Distance depending on altitude and position in the swath.

Altitude/position in the swath	SSD ACT [m]
Max altitude / Centre FoV	~29.8m
Max altitude / Edge FoV	~30.5m
Min altitude / Centre FoV	~28.8m
Min altitude / Edge FoV	~29.7m

5.3 Spatial resolution

The MRD specifies the spatial resolution of the mission with a MTF at Nyquist performance greater than 0.2 as threshold and 0.3 as goal for both ALT and ACT directions. The requirement has been flown down with a specified performance of 0.25 in the SSRD.

The actual performance and resulting PSF of the Instrument is the combination of the different elements in the instrument optical chain, with different contributors to the ACT and the ALT direction. Whilst certain contributors in the chain are of top-hat nature (i.e. Slit, ALT smear or detector sampling) the effective resultant system LSF in each ACT and ALT direction can be modelled by a gaussian shape for the purposes of application analysis.

Thanks to this approximation we can define the full system PSF for all pixels along the swath as the combination of ACT and ALT gaussians.

$$\sigma_{PSF} = \frac{p}{\pi}\sqrt{-2\,ln(MTF_{Nyquist})} \tag{10}$$

$$PSF(x,y) = \frac{1}{2\pi\sigma_x\sigma_y} exp\left(-\frac{1}{2}\left[\left(\frac{x}{\sigma_x}\right)^2 + \left(\frac{y}{\sigma_y}\right)^2\right]\right) \tag{11}$$

where p is the interpixel distance and MTF at Nyquist for each direction is defined in the table below:

Table 3 Across track and Along-track estimated worst case MTF performance including tolerances and ground to orbit and lifetime effects. The proposed parameters are a conservative envelope of the CDR predictions with retained system margin but compliant to the requirement.

Direction	MTF at Nyquist
Across track	~0.25
Along track	~0.28

The MTF formulation above is defined in instrument geometry for the ACT and ALT directions. As such the PSF modelling should be applied into an instrument geometry grid. If however the PSF grid is being applied to an orthorectified grid directly then there are three options to make use of the current modelling:

- Apply as is, this introduces a small approximation error due to the difference in alignment between the orthorectified geometry and the sensor geometry. As the disparity between ALT and ACT expected values is small this approximation error is likely second order for most applications
- Rotate the PSF kernel to align the ACT and ALT principal directions of the PSF appropriately and then apply the kernel. This just requires a simple application of a rotation matrix to the PSF kernel, see figure below
- Transform the grid into sensor geometry before applying the kernel

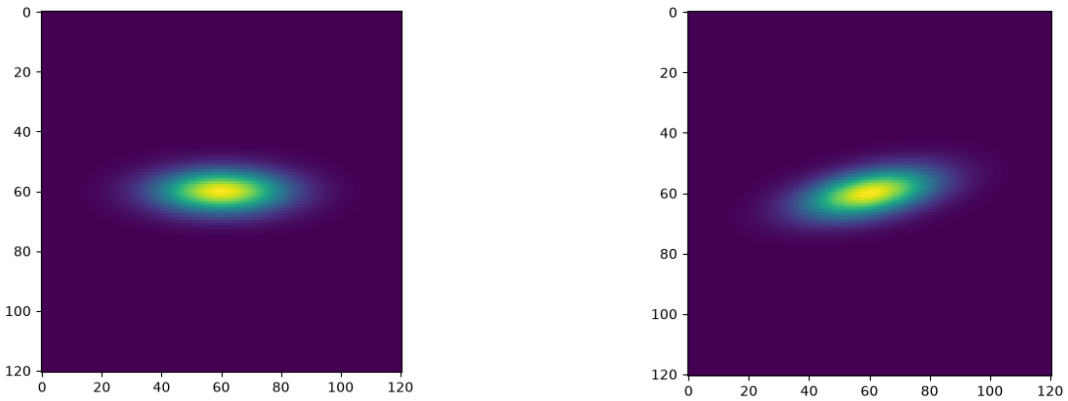


Figure 14. Rotation of PSF kernel before application in case sensor geometry projection is not performed. ESA suggested approach is to transform the grid into sensor geometry and then work in sensor geometry axis instead, see Section 6

In section 6 as part of the full pipeline suggested implementation we propose to implement the third option. This option is the most complete and allows working in sensor geometry which is also useful for other modelling aspects, e.g. equalization error, see Figure 16.

6. SUGGESTED INTEGRATION OF PERFORMANCE MODELLING FOR E2E PERFORMANCE ANALYSIS

The model provided in the sections above simulates the Instrument acquisition for a given at-sensor oversampled spectral radiance. In order to use these equations in an E2E context and perform further application analysis extra aspects are required to be defined to link the geophysical variables with the at-sensor radiances. Moreover, the right spectral and spatial oversampling needs to be considered in the inputs so that the results fidelity is guaranteed.

Below we suggest a simplified architecture implementation for simplified E2E analysis based on the performance model described in the previous sections. This architecture is meant to be taken as a guidance for application analysis but not a mandatory implementation.

The pipeline can be divided into 4 main elements. In the table below we clarify the role of each step and its inputs and outputs.

Table 4 Suggested architecture for simplified pipelines that inject the proposed performance model. I is the across-track index, J along-track and K spectral.

Module Name	Purpose	Input	Output
Scene generation	Defines the input scene, which has to be spatially and spectrally oversampled	NA	BOA reflectance hypercube [Spat_OF I, Spat_OF J, Spec_OF K]
Atmospheric propagation	Propagates the bottom of atmosphere reflectance to Top of Atmosphere Radiance	BOA reflectance hypercube [Spat_OF I, Spat_OF J, Spec_OF K] Atmospheric model	TOA radiance hypercube [Spat_OF I, Spat_OF J, Spec_OF K]
Instrument model	Implements the equations from this paper to simulate L1 acquisitions	TOA radiance hypercube [Spat_OF I, Spat_OF J, Spec_OF K]	Simulated L1 TOA radiance hypercube [I, J, K]
Atmospheric correction	Applies atmospheric correction with a model consistent to retrieve BOA reflectance	Simulated L1 TOA radiance hypercube [I, J, K]	Retrieved BOA reflectance [I, J, K]

Scene considerations

In order to limit error introduced by the spatial and spectral convolutions required in the Instrument model, an oversampling factor between 5-10 spectrally and spatially is recommended, which is a practical guidance range identified adequate for CHIME applications in line with the oversampling factors considered within the CHIME end-to-end simulation activities. Of course this is a tradeoff between processing and memory together with available data at the required spectral and spatial sampling.

In terms of spatial sampling this need leads us to consider airborne campaign imagery as main suggested input. Regarding spectral sampling if the required oversampling is not available, interpolation using spectral libraries to generate the required data is possible, however it can compromise the fidelity of the modelling specially for biomes with sharp absorptions (e.g. mineral biomes).

Atmospheric forward model

Top of Atmosphere spectral radiance needs to be generated from the scene bottom of atmosphere reflectance. Multiple solutions are available for this purpose, either opensource (e.g. 6s, ERADIATE) or commercial (e.g. MODTRAN). What is important is that the key atmospheric aspects are stored so that an ideal atmospheric correction can be later modelled. These are:

- 2 way atmospheric transmittance
- Path radiance
- Spherical albedo

Simplified plots showing these properties are depicted below for reference, the transmittance, path reflectance and spherical albedo are defined from a simplified model for the purposes of demonstration:

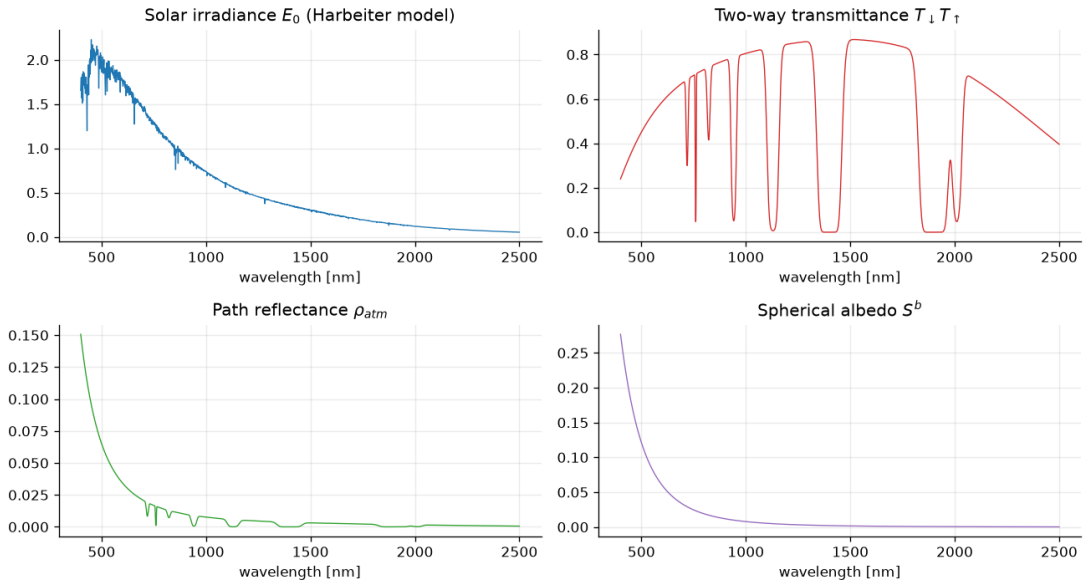


Figure 15. Depiction of the atmospheric properties required for the forward model to be retained to model ideal atmospheric correction.

Instrument modelling

The Instrument acquisition geometry is different to the input orthorectified geometry. Working in sensor geometry is important for two main aspects described in the previous sections:

- The PSF is asymmetrical and has principal directions in the ACT and ALT directions, which are not aligned with the pixels in the ortho geometry
- The equalization residual manifests only in ACT (i.e. striping). Hence to model this effect adequately, the acquisition needs to be in sensor geometry.

For application analysis, large extent areas are very rarely simulated. The typical use case is a local scene where the specific biomes of interest are located over a region of tens of km. This allows us to introduce a key simplification in the chain to change from ortho to sensor geometry. The high-fidelity way of simulating the acquisition is to introduce a direct location model that models each pixel line of sight of the satellite and intersects it with the orthorectified geometry, resample it and convert it into an at-sensor grid geometry. However, for small spatial scales this can be significantly simplified by modelling a simple rotation in the grid from the ortho grid to the ground track. This approximation holds for the scene size just described and allows to keep the pipeline independent from a complex orbit and direct location framework.

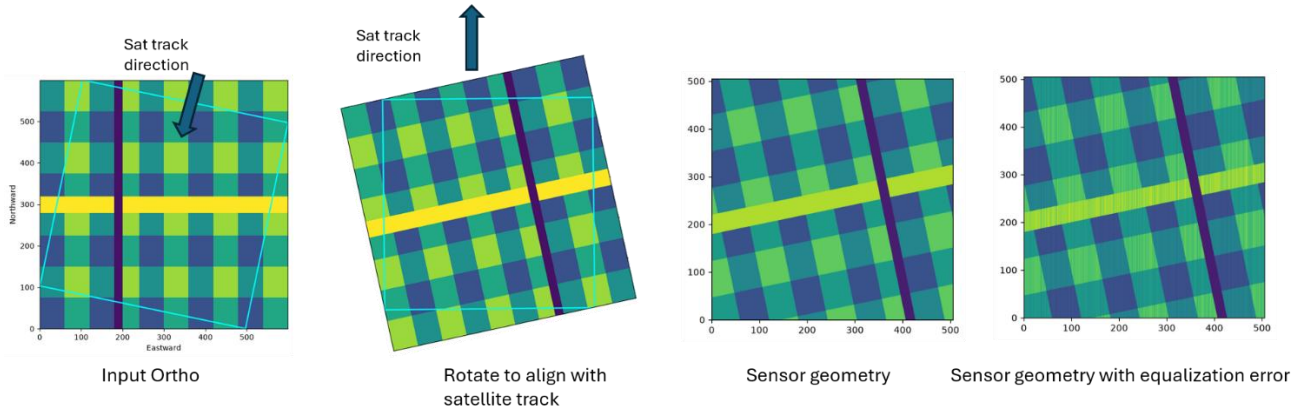


Figure 16. Suggested geometric transformation to model sensor geometry from an orthorectified input. Rotation based on the expected satellite track can be a good approximation if the scene spatial scale is of few 10s of km.

For CHIME orbit the satellite track angle vs the ortho grid varies approximately as depicted in the figure below (derived assuming circular orbit and spherical Earth):

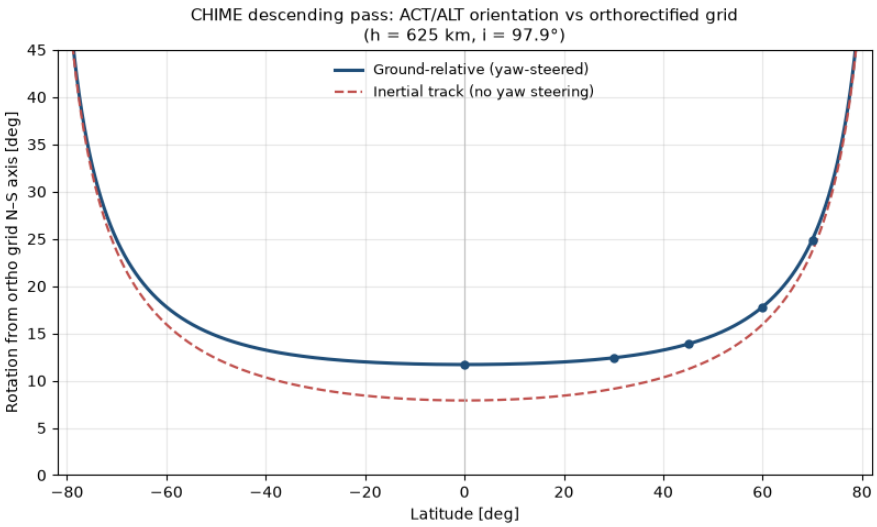


Figure 17. Relative angle between ground track and orthorectified grid, computed following a circular orbit and spherical Earth approximation.

For the majority of areas of interest, the heading angle evolution vs latitude is small and within 10-20 degrees justifying the approximation just discussed.

We suggest performing the implementation of the spectral, spatial and radiometric modelling in the order below:

- Spectral convolution including spectral knowledge error in the spectral grid
- Spectral downsampling from Spec_OF K down to K samples
- Scene rotation to simulate instrument geometry acquisition as per figure
- Spatial convolution
- Spatial downsampling from [Spat_OF I, Spat_OF J] down to [I,J]
- Random addition of temporal noise (SNR) for each pixel as a random draw from normal distribution $N(0,NEDL^2)$
- Addition of radiometric error (ARA) and equalization residual error (RXRA)

A key aspect of the Instrument modelling is the understanding of the different performances error treatment in the chain. Below we provide the description of the nature of each performance error introduced in the chain:

Table 5 Suggested error treatment across simplified chain

Error	Nature	Characteristic timescale	Modelling approach
Temporal noise (SNR)	Radiometric temporal variation	Varies per frame	Gaussian distribution random draw using a standard deviation equal to the computed NEDL= L/SNR independently for each spatial pixel and spectral channel
Radiometric accuracy gain error	Bias and long term drift between Solar and Lunar calibrations	Bias for diffuser knowledge Weeks for cal residual	For users interested in one scene acquisition is a bias at scene level For users interested in temporal acquisitions drift behaviour should be included
Radiometric accuracy offset error	Evolution between dark calibrations	Orbital timescale	For users interested in one scene acquisition is a bias at scene level For users interested in temporal acquisitions a random error modelling is likely more adequate.
Relative Spatial Radiometric Accuracy	Evolution between equalization campaigns	Months	Spatially is modelled as a random draw for each column. Temporally is a bias at scene level and a drift of a timescale of 6 months

Atmospheric correction

If propagation to Bottom of Atmosphere reflectance is required for application analysis then multiple options are possible. The instrument output in spectral radiance can be converted to top of atmosphere reflectance. From the top of atmosphere reflectance (ρ*) either a commercial or open source atmospheric correction model can be utilised; here the atmospheric parameters are determined directly from the product and complementary meteo data (e.g. ECMWF/CAMS) consistent with the atmospheric forward model. Alternatively, the surface reflectance can be recovered assuming Lambertian surface and perfect knowledge of the atmospheric properties using the parameters used in the Atmospheric forward model before and the equations below where ρ_s is the target surface reflectance.

$$\rho^*(\lambda) = \frac{\pi L_{sensor}(\lambda)}{\cos(\theta_{SZA}) E_0(\lambda)} \quad (12)$$

$$y(\lambda) = \frac{\rho^*(\lambda) - \rho_{atm}(\lambda)}{T_{\downarrow}(\lambda) T_{\uparrow}(\lambda)} \quad (13)$$

$$\rho_s(\lambda) = \frac{y(\lambda)}{1 + S^b\, y(\lambda)} \quad (14)$$

Below we provide an example for a single pixel vegetated and mineral biomes for different breakpoints across the chain, including radiometric noise and error but ideal atmospheric correction. Same approach can be expanded to full scene datasets.

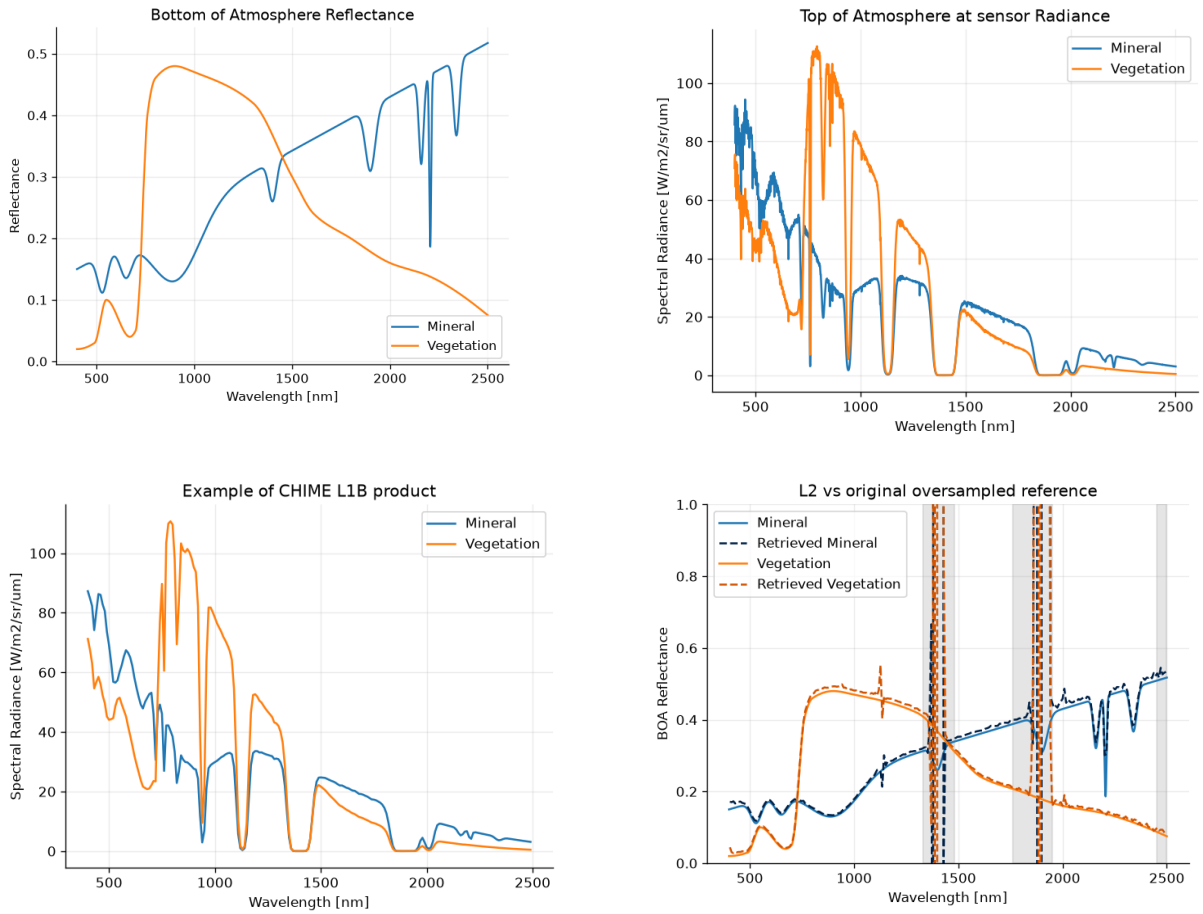


Figure 18. Single pixel vegetation and mineral examples of full pipeline from Bottom of Atmosphere to L2 retrieved reflectance. The effects of a positive gain error of 2.5% and the SNR can be clearly seen in the L2 comparison vs the truth

The full simplified chain, together with the simplified performance models and the data to run them is being currently developed in an opensource GitHub project stored in https://github.com/esa/chime-performance-sup.git. The repository will be setup for the CDR performances and updated once Instrument TVAC characterization figures are available.

7. CONCLUSION AND FUTURE WORK

Satellite system performance is traditionally communicated to the user community through mission and system requirements. While this provides a useful first-order basis for performance modelling, requirements alone are not sufficient to represent the overall mission behaviour needed for application-oriented analysis. They typically describe performance at a limited number of reference conditions and do not provide the granularity required to simulate realistic scenes across the full instrument dynamic range. Conversely, complex end-to-end simulators are developed during the programme lifecycle to support several use cases, ranging from performance verification and processor development to data generation and user preparation. However, these simulation tools are tailored to expert users, contain significant design detail that cannot be shared with the wider community and are computationally demanding.

This paper proposes an intermediate approach specifically targeted at supporting CHIME user preparation activities. We present a set of parametric models that can be readily integrated into simulation pipelines while retaining the level of fidelity required for application analysis. This results in a lightweight, shareable reduced-order bridge between requirement-level reporting and the proprietary/high-fidelity E2E (OPSI/CHEES) framework. The approach captures the dominant radiometric, spectral, spatial, and geometric performance behaviours relevant to users, including SNR and NEDL across the radiometric dynamic range, absolute radiometric accuracy over uniform scenes, additional considerations for heterogeneous scenes, relative spatial radiometric accuracy and residual striping, spectral response and sampling effects, spectral misregistration related considerations, and spatial resolution through a simplified PSF representation. In this way, the proposed model provides more actionable information than requirement-only reporting, while avoiding the operational complexity and sharing constraints of full high-fidelity simulators.

The model and parameters provided in this paper reflect the CHIME System CDR status and will evolve as subsystem manufacturing, integration, and performance characterisation progress, in particular following instrument-level testing and thermal-vacuum characterisation. Future work will focus on consolidating the parameter sets, validating the simplified assumptions against higher-fidelity simulations and measured performance, and making the implementation available through the open repository already identified for this activity (https://github.com/esa/chime-performance-sup.git), so that application teams can integrate the model into their own processing chains. This decentralised approach is expected to improve early application readiness, support more realistic exploitation studies, and provide a useful precedent for communicating pre-launch mission performance to user communities in future Earth observation programmes.

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The models, analyses, and parameter derivations presented in this paper were developed by the authors. Generative AI tools were used during manuscript preparation for code prototyping, plot generation, technical review, text polishing, and as an independent cross-check that the reported models and parameters reproduce equivalent results to the authors' original code prototype. The authors reviewed and verified all AI-assisted outputs and take full responsibility for the content of this publication.

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APPENDIX

Table 1 RXRA channels with performance exceeding 0.4% allocation

Spectral centre	RXRA [%]
750	0.7
760	1.7
900	0.7
910	0.7
920	0.7
930	1.7
940	1
1120	0.7
1130	0.7
1150	0.7
1490	0.7
1500	0.7
1990	0.7
2000	1.7
2010	0.7
2020	0.7
2050	1
2350	0.7
2360	0.5
2380	0.7
2420	1
2430	1.7
2440	1.5
2450	1

Table 2 Lref, SNR and ARA parameters

Wavelength [nm]	Lref [W/sr/m2/um]	SNR g	SNR N0	ARA a	ARA b
404.2	128.90771	298.6249	1.477043	0.000624	89176.85
412.6	133.84654	340.0457	1.139124	1.56E-15	115631.1
421	126.22583	359.3458	1.020047	0.00076	129129.4
429.4	111.04402	362.4488	1.007215	0.000786	131369.2
437.8	121.09816	397.3595	0.868355	0.000239	157894.6
446.2	133.72809	436.3473	0.720112	0.001676	190399
454.6	139.89916	461.5604	0.62953	0.000165	213038
463	138.44668	468.4552	0.602948	4.76E-12	219450.2
471.4	137.5161	476.7573	0.579496	0.001236	227297.5
479.8	135.09071	484.8682	0.560271	0.00154	235097.1
488.2	125.84713	475.3444	0.582946	0.001186	225952.3
496.6	124.18185	477.1839	0.578461	0.000389	227704.5
505	119.2398	471.7763	0.591797	0.000309	222572.9
513.4	114.91228	470.4804	0.595062	4.22E-12	221351.8
521.8	109.8847	465.0446	0.609055	0.001718	216266.4
530.2	111.1564	468.5233	0.600044	3.01E-11	219514.1
538.6	109.61031	463.9843	0.611841	0.001655	215281.4
547	107.67719	463.0329	0.614358	0.001599	214399.5
555.4	105.79908	463.9633	0.611897	1.32E-11	215261.9
563.8	102.66003	463.4917	0.613142	0.001625	214824.6
572.2	100.62135	463.1452	0.61406	0.001604	214503.5
580.6	100.21861	465.2653	0.608477	0.001731	216471.8
589	92.867205	450.5804	0.648785	2.46E-11	203022.7
597.4	91.187679	449.6678	0.651421	4.25E-05	202201.1
605.8	93.487096	459.9831	0.622532	8.33E-14	211584.5
614.2	90.849108	458.2248	0.627319	3.66E-05	209970
622.6	89.566801	459.0853	0.624969	7.31E-13	210759.3
631	87.435994	457.4502	0.629445	0.000412	209260.7
639.4	87.58425	451.9124	0.644966	0.001929	204224.8
647.8	83.239229	437.376	0.68855	0.001638	191297.7
656.2	80.776993	435.9669	0.693008	0.001681	190067.1

Wavelength [nm]	Lref [W/sr/m2/um]	SNR g	SNR N0	ARA a	ARA b
664.6	84.427432	451.8553	0.645129	5.32E-12	204173.2
673	84.317528	456.1324	0.633087	0.000266	208056.8
681.4	80.98135	449.3878	0.652233	0.001658	201949.4
689.8	69.666184	419.5387	0.751747	0.000802	176012.7
698.2	70.978132	436.3834	0.770316	0.001676	190430.4
706.6	72.46006	488.5753	0.712384	0.001694	238705.8
715	65.263978	465.2914	0.785466	0.000191	216496.1
723.4	53.911402	423.7613	0.946967	0.001088	179573.6
731.8	57.953048	438.6577	0.883742	0.001531	192420.6
740.2	66.80525	472.5059	0.761663	6.11E-11	223261.8
748.6	69.886637	480.3061	0.737125	3E-11	230694
757	57.816687	430.1574	0.919015	0.001398	185035.4
765.4	43.035531	370.5402	1.173096	1.77E-12	137300
773.8	63.924045	453.1176	0.828239	0.00103	205315.5
782.2	64.833067	452.5002	0.8305	2.3E-11	204756.4
790.6	60.887326	432.8748	0.907513	0.001622	187380.6
799	58.867111	423.1838	0.949553	0.000902	179084.5
807.4	56.796941	416.0074	0.982596	1.2E-11	173062.2
815.8	46.411478	375.6225	1.141566	3.34E-12	141092.3
824.2	45.028913	369.8734	1.17733	1.4E-11	136806.3
832.6	48.293987	383.0411	1.097776	1.58E-11	146720.5
841	52.97336	398.7215	1.013129	0.000451	158978.8
849.4	52.746645	396.0618	1.026782	5.86E-14	156865
857.8	51.90446	392.467	1.045678	6.03E-12	154030.3
866.2	51.319168	391.5422	1.050624	3.08E-11	153305.3
874.6	51.339018	392.5232	1.045379	0.00017	154074.4
883	50.205796	388.462	1.015753	4.6E-12	150902.8
891.4	45.777489	371.8159	1.16506	0.000887	138247
899.8	33.871624	320.7416	1.565646	0.000529	102875.2
908.2	31.764356	323.0519	1.543333	0.000621	104362.5
916.6	30.032841	314.6576	1.626776	0.000605	99009.41
925	29.806666	314.2396	1.631107	0.000727	98746.5

Wavelength [nm]	Lref [W/sr/m2/um]	SNR g	SNR N0	ARA a	ARA b
933.4	11.752423	197.6402	4.123384	7.24E-05	39061.64
941.8	9.8354928	180.9676	4.918161	0.000195	32749.26
950.2	10.534696	187.6125	4.575943	4.83E-05	35198.45
958.6	14.198953	218.3736	3.377566	0.000416	47687.05
967	23.602258	282.4201	1.921738	1.93E-12	79761.09
975.4	28.81052	312.6378	1.402754	0.000862	97742.38
983.8	33.833571	339.7408	1.141169	0.000685	115423.8
992.2	38.533275	364.0341	0.993943	6.53E-12	132520.8
1000.6	38.690991	366.8596	0.978691	1.1E-11	134586
1009	37.888544	365.4835	0.986075	3.32E-12	133578.2
1017.4	37.596194	366.4384	0.980942	2.18E-11	134277.1
1025.8	37.125914	366.3199	0.981577	0.000128	134190.3
1034.2	36.625586	354.6459	1.047263	2.09E-07	125773.7
1042.6	36.051475	337.0756	1.159287	1.7E-11	113619.9
1051	35.482181	337.659	1.155284	0.000631	114013.6
1059.4	34.696519	337.9378	1.153379	0.000639	114201.9
1067.8	33.492148	335.0586	1.173286	7.51E-12	112264.2
1076.2	32.033482	329.9187	1.210129	4.27E-05	108846.3
1084.6	29.978634	322.0607	1.269901	0.000581	103723.1
1093	26.876013	317.6033	1.305796	0.000817	100871.8
1101.4	21.565033	302.9126	1.539966	0.000599	91756.03
1109.8	14.607212	251.5064	2.54628	8.4E-05	63255.47
1118.2	4.8276673	145.5545	7.602436	3.79E-15	21186.12
1126.6	2.2571322	100.068	16.08475	1.95E-14	10013.6
1135	3.5128414	125.8119	10.17562	1.41E-12	15828.63
1143.4	4.6633443	147.0029	7.453361	1.75E-05	21609.87
1151.8	5.3062105	158.9549	6.374649	0.00021	25266.67
1160.2	11.113047	231.313	3.010262	2.06E-14	53505.69
1168.6	16.965812	286.5731	1.961252	0.000701	82124.15
1177	19.175748	307.4599	1.621468	0.000837	94531.58
1185.4	19.065062	309.8315	1.67785	0.00079	95995.56
1193.8	20.293298	322.2007	1.476496	0.000586	103813.3

Wavelength [nm]	Lref [W/sr/m2/um]	SNR g	SNR N0	ARA a	ARA b
1202.2	19.812777	321.8231	1.555141	2.01E-12	103570.1
1210.6	20.439066	331.1062	1.469162	0.000957	109631.3
1219	21.778434	344.9778	1.353387	2.41E-05	119009.7
1227.4	22.85374	354.4856	1.219799	0.000298	125660
1235.8	24.004165	365.0541	1.100755	7.15E-12	133264.5
1244.2	24.15591	370.455	1.1169	0.00089	137236.9
1252.6	23.614044	370.8443	1.171173	0.000891	137525.5
1261	21.618347	357.7048	1.258794	0.000739	127952.8
1269.4	19.640863	342.4172	1.373703	1.78E-11	117249.6
1277.8	21.065983	355.8988	1.271602	6.37E-12	126664
1286.2	20.951501	356.1248	1.269988	3E-12	126824.9
1294.6	19.63209	345.9382	1.28082	0.000358	119673.3
1303	16.947764	323.6584	1.537555	3.93E-12	104754.7
1311.4	14.467143	301.9157	1.766985	0.000508	91153.09
1319.8	12.030093	277.9768	2.08443	5.56E-15	77271.09
1328.2	8.2957249	232.3734	2.982849	2.5E-12	53997.42
1336.6	5.1440198	184.0024	4.757264	3.88E-12	33856.89
1345	2.061288	117.1818	11.72961	8.87E-06	13731.58
1353.4	0.1689801	33.71496	141.6964	1.62E-06	1136.699
1361.8	0.0594359	20.01462	402.0772	5.8E-07	400.5851
1370.2	0.0593932	20.07527	399.6514	2.93E-07	403.0166
1378.6	0.0622185	20.62882	378.4908	4.82E-07	425.5483
1387	0.0597522	20.36365	388.4125	2.32E-05	414.6781
1395.4	0.0610798	20.72853	374.8584	3.1E-05	429.672
1403.8	0.0633327	21.17378	359.2589	4.16E-05	448.3288
1412.2	0.0884859	25.05315	256.6136	1.27E-14	627.6601
1420.6	0.163512	34.12406	138.3194	8.4E-06	1164.451
1429	0.359533	50.82004	62.36405	2.22E-05	2582.676
1437.4	0.3627277	51.33396	61.12159	3.56E-06	2635.176
1445.8	0.6968553	71.52266	31.48596	4.32E-05	5115.491
1454.2	1.6372728	109.963	13.32021	8.6E-05	12091.87
1462.6	2.0958082	124.6556	9.864202	8.32E-05	15539.01

Wavelength [nm]	Lref [W/sr/m2/um]	SNR g	SNR N0	ARA a	ARA b
1471	1.3257284	99.55699	15.46471	7.95E-05	9911.594
1479.4	2.0846789	125.5353	9.308373	2.39E-14	15759.1
1487.8	3.8411719	167.8867	5.013174	2.34E-12	28185.96
1496.2	7.279002	218.6324	2.7556	0.000411	47800.14
1504.6	9.4513295	248.1167	2.139605	0.000513	61561.88
1513	11.304491	271.846	1.782377	0.000541	73900.23
1521.4	12.755895	290.0148	1.573169	8.47E-13	84108.58
1529.8	13.677008	301.5316	1.448704	0.00047	90921.32
1538.2	14.127916	307.1567	1.396129	0.000832	94345.22
1546.6	14.307971	316.1176	1.318099	8.34E-12	99930.36
1555	14.24542	336.2693	1.424393	3.12E-12	113077.1
1563.4	13.973374	335.865	1.427825	3.24E-12	112805.3
1571.8	12.656166	319.7706	1.575169	9.55E-06	102253.3
1580.2	12.556828	318.3078	1.58968	1.57E-11	101319.8
1588.6	13.040602	323.3578	1.540415	0.000633	104560.2
1597	12.814545	319.5181	1.501392	0.000962	102091.8
1605.4	11.944164	308.9942	1.686955	0.000828	95477.43
1613.8	12.192758	314.0282	1.633304	0.000629	98613.71
1622.2	12.720926	321.9443	1.478849	1.74E-12	103648.1
1630.6	12.647426	321.3655	1.48418	0.000553	103275.8
1639	11.867663	311.6834	1.577822	0.000601	97146.53
1647.4	11.595679	308.6946	1.608523	0.000835	95292.37
1655.8	11.710693	311.0378	1.584379	0.000686	96744.49
1664.2	11.385658	307.1775	1.62445	0.000832	94358.01
1672.6	11.230486	305.0937	1.575937	0.00076	93082.16
1681	10.908801	300.6631	1.622725	0.000764	90398.31
1689.4	10.770496	298.7782	1.582879	0.000695	89268.41
1697.8	10.474116	294.8917	1.624877	0.000191	86961.12
1706.2	10.012548	288.3275	1.699704	8.15E-13	83132.77
1714.6	9.7682002	284.5245	1.745445	0.000678	80954.21
1723	9.0913206	273.9827	1.882346	0.000566	75066.5
1731.4	8.5408866	265.4087	2.005928	3.43E-12	70441.77

Wavelength [nm]	Lref [W/sr/m2/um]	SNR g	SNR N0	ARA a	ARA b
1739.8	7.886832	255.3423	2.167206	0.000388	65199.67
1748.2	7.5787549	250.5289	2.251281	7.07E-05	62764.75
1756.6	7.2382975	245.2461	2.349316	0.000445	60145.64
1765	6.1481486	226.3224	2.86385	0.000251	51221.81
1773.4	4.8780928	201.5347	3.773859	3.75E-05	40616.22
1781.8	3.2240088	163.5089	6.024506	1.33E-05	26735.16
1790.2	2.0304465	129.2223	9.645606	1.16E-12	16698.39
1798.6	0.6752458	74.2724	29.19775	5.88E-17	5516.389
1807	0.1268483	32.14266	155.898	8.85E-14	1033.151
1815.4	0.0264614	14.68867	746.5159	2.05E-07	215.7571
1823.8	0.0146261	10.91702	1351.438	5.44E-15	119.1813
1832.2	0.0113118	9.627999	1737.529	2.79E-05	92.69835
1840.6	0.0094829	8.807263	2076.444	3.8E-16	77.56807
1849	0.0094992	8.792938	668.5355	1.89E-14	77.31576
1857.4	0.009485	8.773129	671.5579	7.52E-21	76.96779
1865.8	0.0094287	8.76174	673.3048	1.3E-06	76.7681
1874.2	0.0077553	7.946138	818.615	5.71E-15	63.14116
1882.6	0.0100839	9.058445	629.9196	1.66E-10	82.05543
1891	0.0101501	9.058021	629.9786	1.04E-14	82.04775
1899.4	0.0086726	8.371261	737.5826	4.45E-15	70.07801
1907.8	0.0084119	8.242365	760.8318	9.65E-07	67.93659
1916.2	0.0091448	8.585081	701.2994	2.1E-14	73.70363
1924.6	0.0104196	9.161344	615.8489	1.02E-14	83.93022
1933	0.0142004	10.67739	453.3796	4.93E-16	114.0067
1941.4	0.0453834	19.04926	142.4414	2.37E-05	362.8742
1949.8	0.1304576	32.30085	49.54097	4.37E-14	1043.345
1958.2	0.2394931	43.85127	26.87993	3.21E-05	1922.934
1966.6	0.9803433	88.79086	6.556258	1.24E-13	7883.816
1975	2.0960485	129.659	3.074589	2.9E-14	16811.46
1983.4	3.2234099	160.2937	2.044101	3.04E-12	25694.08
1991.8	3.2618137	160.8342	2.030387	0.000141	25867.63
2000.2	1.4999002	108.8434	4.363033	0.000105	11846.88

Wavelength [nm]	Lref [W/sr/m2/um]	SNR g	SNR N0	ARA a	ARA b
2008.6	0.8208035	80.5969	7.957118	2.1E-13	6495.861
2017	1.0561574	91.61968	6.15765	1.52E-15	8394.165
2025.4	2.7759189	148.6474	2.376952	7.35E-06	22096.06
2033.8	4.3517302	186.0952	1.516579	0.000228	34631.42
2042.2	4.3563684	185.6359	1.524092	7.15E-12	34460.69
2050.6	3.1504605	157.2399	2.124269	1.05E-14	24724.4
2059	2.6516727	143.945	2.534791	1.54E-12	20720.17
2067.4	2.7914306	147.6205	2.410138	2.05E-12	21791.81
2075.8	3.6626814	169.0912	1.836934	1E-05	28591.82
2084.2	4.384921	184.9237	1.443939	1.96E-12	34196.78
2092.6	4.5959378	188.889	1.30995	7.41E-13	35679.06
2101	4.6592127	189.4243	1.256566	0.000251	35881.58
2109.4	4.8373189	192.5093	1.150461	1.31E-12	37059.82
2117.8	4.6934714	189.1329	1.191903	5E-05	35771.27
2126.2	4.688602	189.0432	1.177203	0.000249	35737.34
2134.6	4.8617199	192.5308	1.107805	0.000273	37068.12
2143	4.8354752	192.469	1.108518	7.11E-05	37044.31
2151.4	4.4179181	184.3481	1.26429	0.000216	33984.23
2159.8	4.2682817	181.4462	1.327967	0.000191	32922.72
2168.2	4.1396673	178.2232	1.364061	1.06E-12	31763.52
2176.6	4.1649532	178.1223	1.32598	9.13E-13	31727.56
2185	3.9326531	172.5578	1.421849	3.23E-13	29776.2
2193.4	4.0018042	173.657	1.363779	2.98E-12	30156.75
2201.8	3.7379377	167.411	1.485385	0.000141	28026.44
2210.2	4.0156388	173.1622	1.369487	0.000218	29985.14
2218.6	4.052094	173.8413	1.358809	3.07E-12	30220.78
2227	4.0213086	173.3703	1.366202	0.000222	30057.24
2235.4	3.9474181	171.9603	1.388698	0.0002	29570.34
2243.8	3.850074	170.1467	1.418459	3.84E-12	28949.91
2252.2	3.7256428	167.2714	1.467644	5.51E-15	27979.72
2260.6	3.564744	163.3414	1.539118	0.000192	26680.4
2269	3.5151127	161.7741	1.569084	1.2E-12	26170.86

Wavelength [nm]	Lref [W/sr/m2/um]	SNR g	SNR N0	ARA a	ARA b
2277.4	3.444568	159.8215	1.607658	1.28E-12	25542.91
2285.8	3.3717916	157.6032	1.653234	0.000161	24838.76
2294.2	3.219891	154.0065	1.731355	0.00021	23717.99
2302.6	3.0609992	150.1957	1.820326	0.000177	22558.75
2311	3.1720906	153.1356	1.751103	0.000204	23450.52
2319.4	2.6293765	139.5789	2.107776	3.22E-12	19482.27
2327.8	2.7901779	143.7388	1.987541	3.9E-12	20660.84
2336.2	2.6255987	139.3011	2.116191	0.000153	19404.81
2344.6	2.2144061	127.7834	2.523532	5.83E-06	16328.59
2353	1.9783928	120.4714	2.868396	1.11E-06	14513.35
2361.4	2.2786304	129.0332	2.466387	2.51E-13	16649.56
2369.8	1.7414796	112.7625	3.290061	9.09E-13	12715.38
2378.2	1.5576391	106.5152	3.708089	5.56E-05	11345.49
2386.6	1.3811227	99.82302	4.22194	3.58E-13	9964.636
2395	1.5884913	106.5654	3.616024	5.74E-05	11356.19
2403.4	1.4436801	101.127	4.015409	6.58E-06	10226.67
2411.8	0.9852164	83.41978	5.982309	3.36E-05	6958.86
2420.2	0.8588256	77.76328	6.884268	4.28E-05	6047.128
2428.6	1.228743	92.93568	4.754437	1.49E-12	8637.041
2437	0.9451132	81.5178	6.179583	4.65E-05	6645.152
2445.4	0.7050357	70.40966	8.333951	2.84E-06	4957.52
2453.8	0.4873126	58.52689	11.98818	2.99E-05	3425.397
2462.2	0.6860437	69.40333	8.525178	2.51E-06	4816.822
2470.6	0.3391716	48.70604	17.31006	1.81E-05	2372.278
2479	0.1494541	32.25652	39.46654	1.02E-13	1040.483
2487.4	0.0264175	13.53835	224.0437	2.66E-05	183.2868
2495.8	0.0185337	11.32847	319.9791	6.12E-16	128.3342